

CS & IT ENGINEERING



Operating System

Memory Management

Lecture - 2

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Recap of Previous Lecture



Topic

Memory Management

Topic

Memory Management Technique



Topics to be Covered



Topic

Contiguous Memory Management Technique

Topic

Paging



Topic : Memory Management Techniques

Contiguous

Non-contiguous

fixed partition

variable partition

suffers $\Rightarrow$ Internal fragmentation

suffers $\Rightarrow$ external fragmentation



Topic : Contiguous Memory Management

- Entire process should be stored on consecutive memory locations



Topic : Fixed Partition Contiguous MMT

helps in protection

m.m. is divided into fixed no. of partitions (sizes may vary);
each partition can accommodate at most one process.

ex:-



⇒ max degree of multiprogramming is limited by
no. of partitions

first process arrives ⇒ P1
size = 90 MB

which partition should be given to P1
↓
decided based on partition allocation policy



Topic : Partition Allocation Policy

1. First fit $\Rightarrow$ search a free partition from starting which can store process
2. Best fit $\Rightarrow$ search the smallest possible partition —||—
3. worst fit $\Rightarrow$ —||— largest —||—
4. Next fit $\Rightarrow$ search a free partition from last allocated partition



Topic : Partition Allocation Policy

ex:-

mm



2 processes arrive

P_1 (size = 120MB) P_2 (size = 70MB)

First fit

150MB allocated

100MB allocated

Best fit

150MB allocated

70MB allocated

worst fit

300MB —||—

150MB —||—

Next fit

150MB allocated

80MB allocated

⇒ fixed partition contiguous mmt. suffers from "Internal fragmentation"

if space allocated to process is larger than the space required by process;
then the extra space which is wasted (not utilized);
is called as internal fragmentation.

ex:- process of size 120MB allocated to a partition of 150MB.
↓
size of internal fragmentation = $150 - 120 = 30 \text{ MB}$



Topic : Variable Partition Contiguous MMT

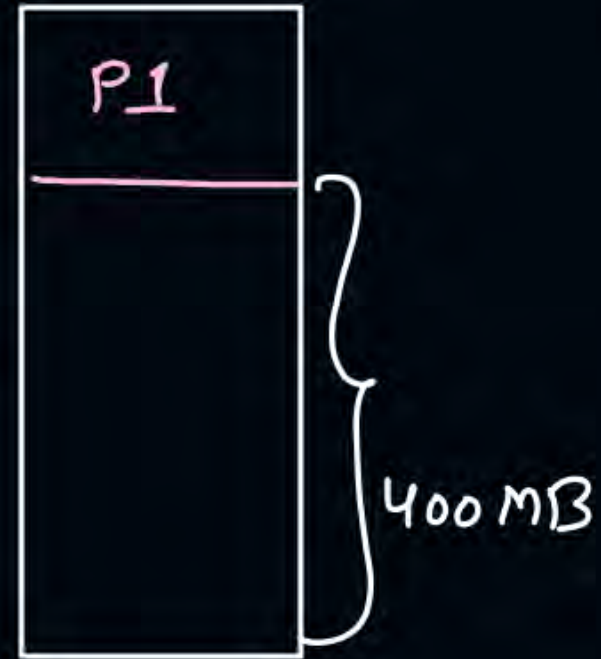
whenever a new process arrives then a new partition is created of the size equal to process size; and the process is allocated into it.

⇒ It does not suffer from internal fragmentation

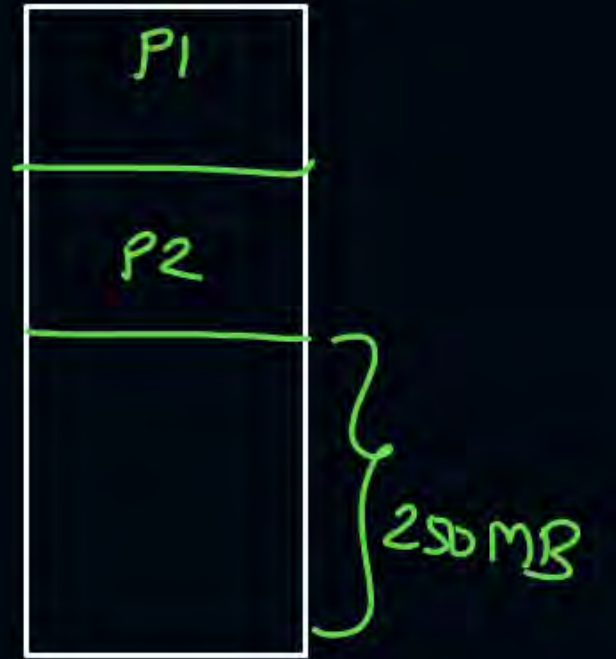


New process
P1
size = 200 MB

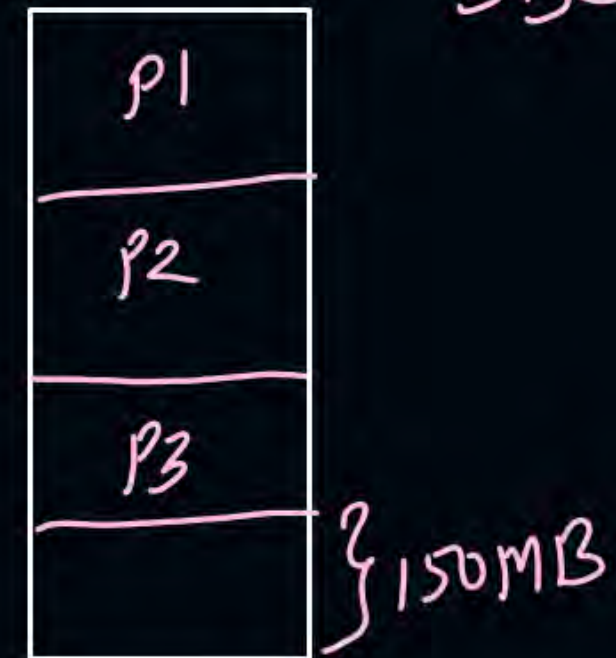
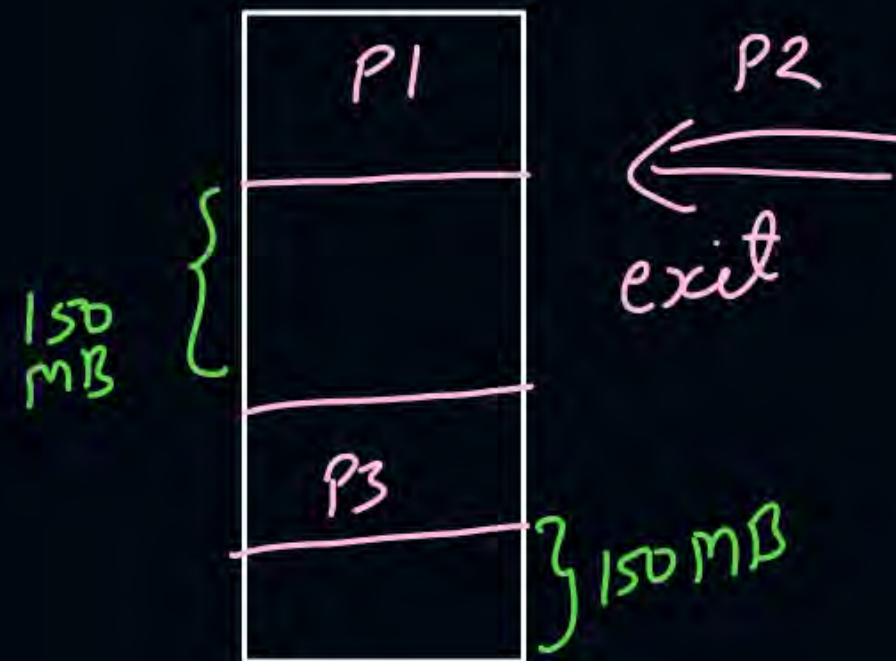
→



New process P2
size = 150 MB



⇓ New process P3
size = 100 MB



New process P4

size = 200MB

P4 Can not be stored in m.m. because 200MB space is not available consecutively.

external fragmentation

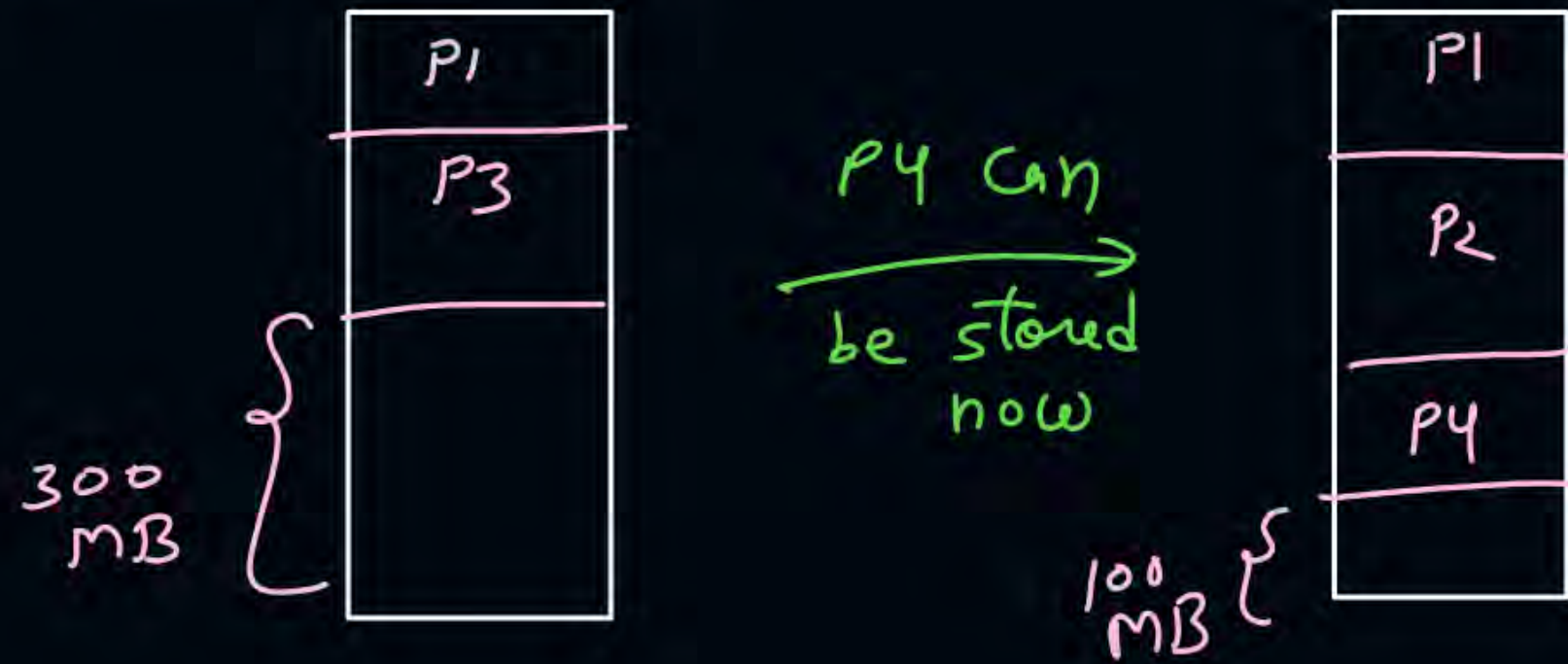
↓

solⁿ :- Compaction

↓

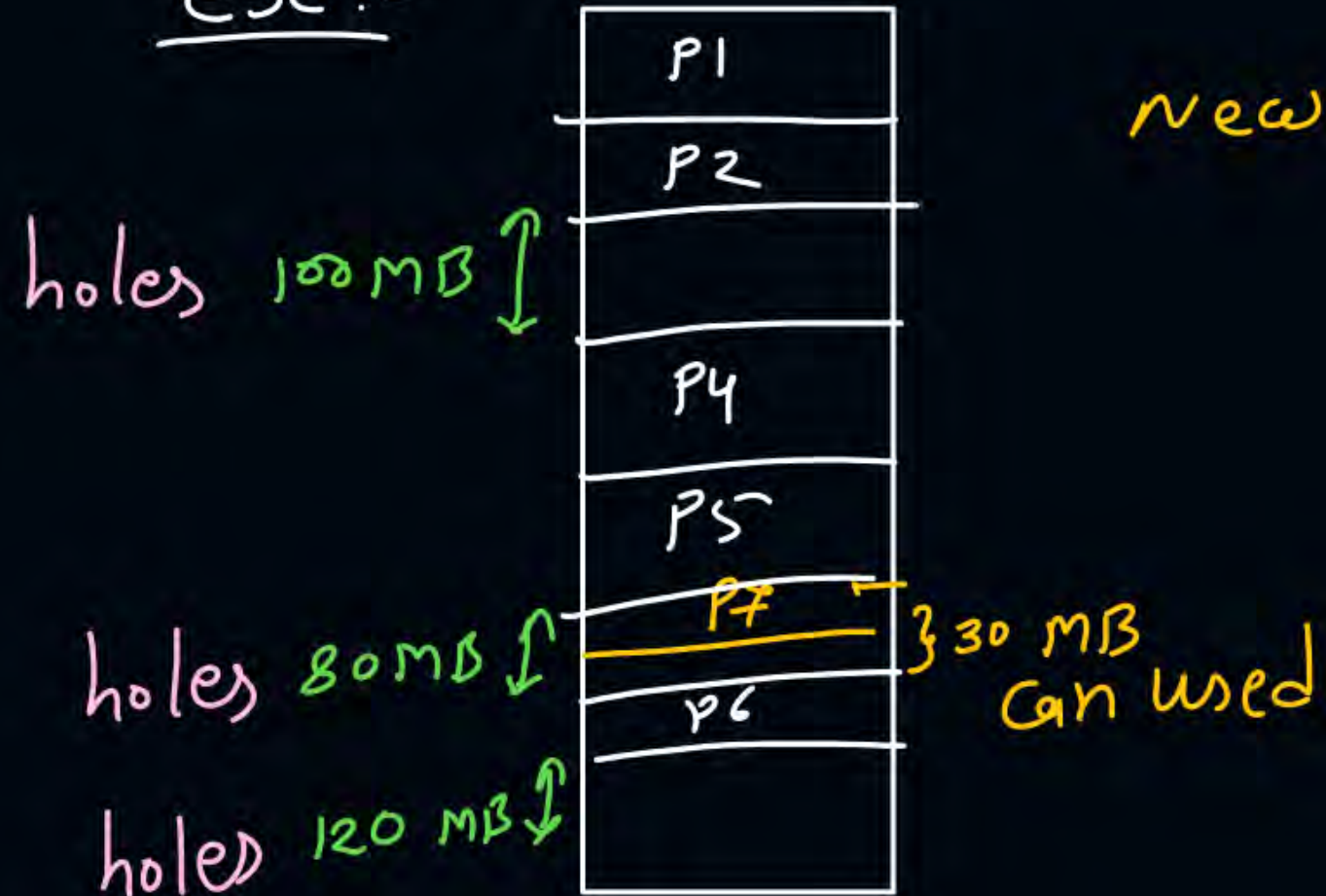
collect all allocated processes into one side of memory, so that entire free space can be in another side together.

After Compaction:-



Partition allocation policy in variable partition contiguous m.m.t.:-

Ex:-



new process of size 50 MB
P7

fixed partition $\Rightarrow$ best fit works best
variable ---||--- $\Rightarrow$ worst fit ---||---

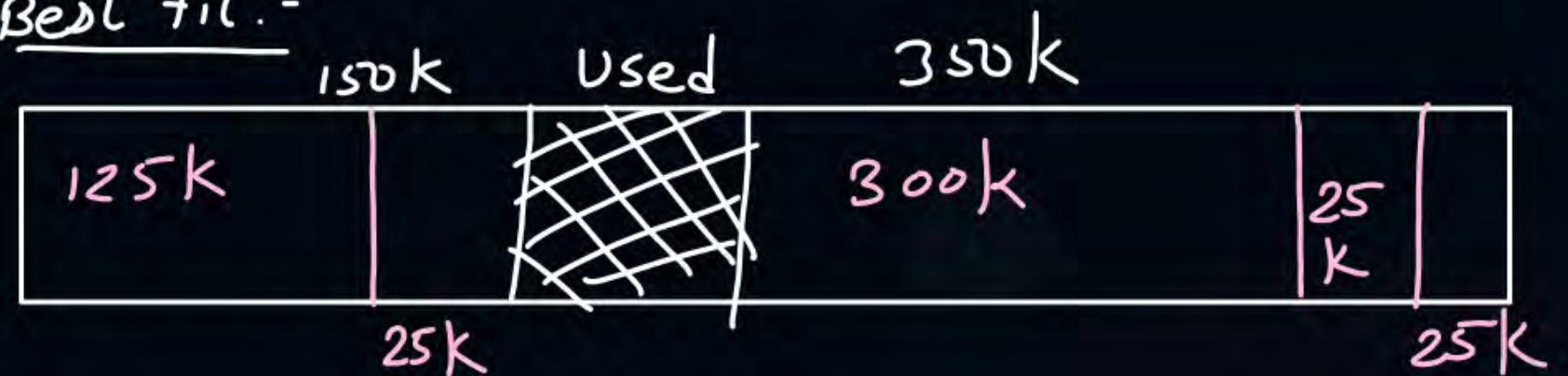
[MCQ]



#Q. Consider the requests from processes in given order 300K, 25K, 125K, and 50K. Let there be two blocks of memory available of size 150K followed by a block size 350K. Which of the following partition allocation schemes can satisfy the above requests? *variable partition mmt*

- A** Best fit but not first fit
- B** ~~First fit but not best fit~~
- C** Both First fit & Best fit
- D** neither first fit nor best fit

Best fit:-



First fit:-



#Q. Consider a fixed partition MMT where there are 5 partitions of size 100MB, 250MB, 200MB, 500MB and 300MB. All Partitions are initially empty. The following process requests are made in the given order:

Process	Size
P1	150MB
P2	400MB
P3	270MB
P4	180MB
P5	80MB

Provide the following answers for First fit, Best fit and Worst Fit policies?
Maximum degree of multiprogramming?
What is the total internal fragmentation size?

#Q. Consider variable partition MMT where there are 4 partitions of size 250MB, 200MB, 500MB and 400MB. The following process requests are made in the given order:

Process	Size
P1	150MB
P2	400MB
P3	270MB
P4	180MB
P5	80MB
P6	50MB

Provide how the processes are stored for First fit, Best fit and Worst Fit policies?



2 mins Summary

Topic

Contiguous Memory Management Technique

Topic

Paging



Happy Learning

THANK - YOU